

Introduction to Industrial Robotics and Automation



Introduction to Industrial Robotics and Automation

Introduction

Robots had virtually no place in our society not too long ago and were confined to the silver screen. Nonetheless, within the last several years, a significant transformation has occurred. Nowadays, robots do a lot of jobs that were formerly done by people, such as performing surgery and assembling items like cars, appliances, and gadgets. Some claim that this is a terrible thing since jobs are being taken by bots and humans are being replaced. As a matter of fact, "when one door shuts, another opens." This is because humans are still needed for the construction, programming, and control of robots. more precisely, individuals who are knowledgeable about industrial automation and robotics.

Manufacturing has been revolutionized by industrial robotics and automation, which have increased accuracy and efficiency. Programmable robots take the place of human labor in jobs like welding and assembling. Because they have sensors, they can securely coexist with people in changing surroundings. Automation reduces manual labor by streamlining processes through the integration of technologies. When combined, they build "smart factories," where equipment talks to each other to produce goods more quickly and with higher quality. Industry 4.0 is here, and it's changing the production environment with this technological transformation.

In this course, you will gain knowledge of the specifics of industrial robots, automation, robot programming, and integration. One of the most important things which is covered in this course as well is safety. You will be able to study the emerging field of industrial robots in the last section. Case studies will demonstrate how different businesses are using robotics and automation to run their operations profitably.

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Module One: An Overview of Industrial Robotics and Automation

The topic of robotics has no disinterested party. Whether used in production or our homes, they can mimic some human activities, penetrate dangerous environments, be launched into space, or are just for fun, robots never cease to amaze and inspire us. Industrial robots, or mechatronic human employees, are the focus of this discussion since they are intended to take the place of humans in particular industrial manufacturing jobs.

Lesson-01: Evolution of Industrial Robotics

Industrial robots have transformed numerous industries significantly. Compared to humans, robots can perform activities faster, more precisely, and accurately by automating them. They have taken over jobs that are hazardous, repetitive, and dangerous, which has greatly increased worker safety in many industrial settings. Over time, industrial robots have evolved in tandem with the ways that manufacturing has been altered.

The Beginning:

In 1937, the first industrial robot was discovered. It was a five-axed crane-like device used for stacking blocks. Several years later, in 1962, UNIMATE became the first industrial robot that a large manufacturer installed. GM automated material management, spot welding, and die casting with UNIMATE. Large, heavy, inflexible devices were mostly employed in the early years of industrial robotics to automate extremely repetitive and straightforward operations. Due to their size and expense, the initial industrial robots were largely employed by big automakers and were quite costly.

A greater number of businesses started producing robots in the late 1960s and early 1970s, which prompted the creation of increasingly sophisticated machinery. Around this period, the first electric robot controlled by a microprocessor was created, as well as the 6-axis robot. Additionally, producers started creating robots tailored to particular uses, like automated arc welding robots.

Advancement of Robotic Technology:

While robotics was beginning to progress by the end of the 1970s, significant advancements in the field did not come about until the 1980s and 1990s. Since new robots and improved controller technology were developed regularly during the 1980s, the decade is recognized as the start of the modern era of robotics. To reduce maintenance, robot designs got increasingly intricate and robust. Beyond the automobile industry, robots also started to automate other industries.

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Present Industrial Robots:

The industrial robots of today are far more sophisticated than those of the past. Robotics have become increasingly autonomous with the development of technologies like vision systems, force sensors, and offline programming. These systems can now operate entire industries on their own. They can automate a greater range of applications and more complicated ones. Very few production-related tasks are too complex for robots to automate. They have also spread to sectors of the economy that are not often associated with industry. Electronic assemblies can be automated with the FANUC Lr Mate 200ic. Although it is possible to use the ABB IRB 1200 Cleanroom for food processing.

Compared to earlier models, modern robots are far more adaptable and reasonably priced. Not just big manufacturers are using robots to automate tasks. Even small and medium-sized operations are being automated by robots. They can work with a high mix of parts and manage small and medium production volumes. More businesses are now able to consider robotic automation because of the advancements in industrial robots.

Lesson-02: Importance Of Automation in The Industrial Sector

Many production operations can be automated through the use of industrial automation systems that incorporate a variety of technologies, such as improved sensors, machine vision systems, and industrial robots. A few procedures where industrial automation is crucial are assembly, material handling, quality control, and packaging. We'll talk about industrial automation today, including what it is and why manufacturing needs it so badly.

Industrial Automation:

The use of cutting-edge technologies to regulate and automate specific manufacturing processes is known as "industrial automation." Automation technology is a crucial component of contemporary production since it may help increase productivity, lower costs, and improve product quality. It can also reduce the risk of workplace accidents by making the manufacturing environment safer. Automation technology includes sensors, computerized control systems, and other tools for operating and controlling machinery and equipment in industrial settings.

Benefits of Industrial Automation:

Because industrial automation provides many advantages for both enterprises and the overall economy, it has gained importance in recent years. The manufacturing environment may be safer, more productive, and more economical with automation. These are the top four advantages of automation, albeit there are many.

- ✓ **Increases Productivity:** The ability of industrial automation to increase output is one of its main benefits. Businesses may manufacture goods and services more quickly, more accurately, and consistently when they use automation. This translates into faster

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production times and higher output since industrial jobs can be completed at a far higher degree than what a human worker could.

Furthermore, automation is capable of continuous operation, enabling production to occur around the clock. This can save expenses and boost output, enabling companies to maintain their competitiveness in the global market.

- ✓ **Improves Quality Control:** Industrial automation can aid in enhancing customer satisfaction and quality control by automating processes like testing and inspection. Businesses can closely monitor and regulate the production process to ensure that products match customer expectations and are consistently of high quality by utilizing sensors and other automation technologies.
This not only improves the reputation of the brand but also fosters consumer loyalty, which is important in markets with intense competition.
- ✓ **Enhances Workplace Safety:** Businesses can lower the risk of mishaps and injuries at work by implementing automation. Machines can do physically taxing or perhaps dangerous jobs that human workers might find too hard to perform. Decreasing the likelihood of workplace accidents and injuries safeguards employees' health and well-being and lowers the danger of legal liability for enterprises.
- ✓ **Reduces Costs:** Automation can have a high upfront cost, but over time, the savings from lower labor costs and higher productivity can add up to a sizable profit. Businesses can lower the energy and raw material requirements for producing items, as well as the amount of waste produced during production, by automating manufacturing operations. In the end, this can reduce costs during the production process in addition to improving environmental sustainability—a problem that both customers and businesses are becoming more and more aware of.

All things considered, industrial automation technology is significant for a wide range of industries and can raise a company's profitability and competitiveness in the world market. We may anticipate seeing an increase in the usage of automation in product production as companies look to take advantage of its many advantages as technology develops.

Lesson-03: Types of Industrial Automation

Industrial automation technologies come in a wide variety, and these can include the following:

- ✓ **Robotic:** Robotic automation is the process of utilizing robots to carry out human-performed jobs. These robots can be taught to carry out a variety of tasks, from straightforward assembly to intricate manufacturing operations. Industries including electronics, pharmaceuticals, and car manufacturing frequently use robotic automation.
- ✓ **Fixed:** Fixed automation, sometimes referred to as hard automation, is the process of carrying out a given operation or set of tasks using specialized equipment. Usually made to create a specific product, these machines can be costly to adjust or reconfigure. In high-

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volume manufacturing lines producing goods like electronics, appliances, and automobiles, fixed automation is frequently employed.

- ✓ **Programmable:** Programmable automation, sometimes referred to as soft automation, uses computer-controlled machinery to carry out tasks. These devices are a more adaptable automation choice because they are simple to reprogram based on the work at hand. For small-scale production lines that need to alter the manufacturing process frequently, this kind of automation is perfect.
- ✓ **Flexible:** Combining fixed and programmable automation results in flexible automation. It entails the use of customized machinery that is easily reprogrammed to carry out a range of functions. For manufacturing lines that create several goods or where it is necessary to alter the production process regularly, this kind of automation is perfect.
- ✓ **Integrated:** A comprehensive system that integrates different forms of automation to optimize the production process is called integrated automation. It consists of devices, software, and sensors that cooperate to operate the production line. Industries that need a high degree of accuracy and precision, such as the manufacturing of aerospace and medical devices, use this kind of automation.
- ✓ **Artificial Intelligence (AI):** AI automation optimizes the manufacturing process through the application of sophisticated algorithms and machine learning. It can be used to find inefficiencies, forecast maintenance requirements, and boost productivity all around. Though it is still in its infancy, this kind of automation has the power to completely transform the manufacturing sector.

Lesson-04: Suitable Industries for Automation with Robots

Automation keeps increasing production efficiency while using fewer resources overall. Even though automation affects many industries, some will—and already do—see phenomenal development due to the efficiency it provides for a variety of corporate procedures. We must look at what makes automation so crucial to comprehend which industries stand to gain the most from it, how it affects organizations, and what future developments we may expect as consumers.

- ✓ **Food industry:** Automation has been more prevalent in the food business in recent years. This has been influenced by rising consumer demand, more regulations about worker and food safety, and more. Because food products are frequently fragile and more challenging to handle directly, automation in the food sector is more sophisticated than in other industries. Furthermore, hygiene is a crucial component. Among other things, the introduction of cobots and improved gripper technology have made things simpler.
- ✓ **Plastics industry:** High-mix low-volume production, which produces a wide range of goods in small quantities, is a common practice in the plastics business. Automation of this is generally challenging due to the constant changes in the production process. This

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has altered due to the adaptability of collaborative robots and advancements in the Smart Industry.

- ✓ **Packaging industry:** In the packaging sector, packing tasks like packing boxes or arranging boxes on pallets for shipping are common. Because the packages are frequently heavy, these are usually monotonous jobs that don't call for much inventiveness or are physically taxing on the staff. Ideal for streamlining, relieving labor, and boosting output.
- ✓ **Electronics industry:** Chips and PCBs are examples of the tiny goods that are used in the electronics sector. After that, these are frequently automatically examined, tested, or even soldered. Little, delicate items that require careful handling. The best robots for this are collaborative ones that use the most sophisticated sensors.
- ✓ **Pharmaceutical industry:** Big numbers and the narrowest error margin are hallmarks of the pharmaceutical sector. Inspection is therefore very important. Because of these qualities, it's the perfect industry for automation. Due to their high accuracy, robots make automation cost-effective and error-prone, especially in large quantities.
- ✓ **Automotive industry:** One of the earliest sectors to use automation was the automotive sector. For years, whole automobile assemblies have been performed by large industrial robots. The Smart Industry has gained interest in this industry due to advances including smaller series sizes and a growing desire for personalization.
- ✓ **Metal industry:** Customized and custom-made products are in high demand in the metal business as well. This can be effectively mechanized because of the introduction of new technologies like Smart Industry and collaborative robots. Press brakes, CNC machines, welding, soldering, and loading are among the automated tasks in this business.

Automation and industrial robots are revolutionizing manufacturing, enabling faster and more efficient processes. Tasks are completed by robots and automation streamlines operations. Smart factories function similarly to extremely intelligent machine teams. Industry 4.0 is the name given to this new manufacturing paradigm. It's similar to a major factory overhaul, making them extremely sophisticated and networked. Ultimately, industrial automation and robotics are paving the way for even more exciting things in the future.

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Module Two: Industrial Robots

Robotics is the study of the design, manufacture, and use of robots as well as computer systems for information processing, sensory feedback, control, and operation of robots. These technologies deal with automated machines that are similar to people in terms of behavior, appearance, and/or mental processes, or that can replace workers in hazardous settings or during production operations.

Robotics has advanced due to the desire to emulate human intelligence and ability as well as the need for automated systems that can finish jobs without human assistance. As a result, a wide variety of robots have been created, from simple machines that can perform tedious tasks to incredibly complex machines that can think and learn.

Lesson-01: Basics of Industrial Robots

An industrial robot is an autonomous system of sensors, controllers, and actuators installed on an articulated frame that carries out specific duties in a manufacturing or processing line. They move continuously and repeatedly guided by a set of commands called a program. These technologies increase processing quality, capacity, and speed while reducing or eliminating the need for human intervention.

It's important to distinguish collaborative robots, a more recent robotic technology, from conventional industrial robots. Collaborative robots, also known as cobots, work side by side with a human operator. They are safe to use in close quarters with human operators because of their restricted force output and the maximum speed limit for cobot operation. This, along with more precise motor current detection, allows them to stop if they come into touch with an item or an operator.

The arm of an industrial robot is its main structural element. The arm's structure is made up of joints and ligaments. Links are the rigid components that move through space inside the robot's grasp. On the other hand, joints are mechanical parts that connect two links and allow for either prismatic or rotational movement. These two components are used to classify the many types of industrial robots.

The most important part of the robot is the end-of-arm tool or end effector. The EOAT is the component that moves or orients the process or product. Specialized jobs that they perform include drilling, cutting, welding, measuring, marking, painting, cleaning, and other related tasks.

Working with a manufacturer or integrator is one way a process owner might put an industrial robotic solution into practice. Industrial robotics manufacturers offer their line of robots under their brands. They are typically referred to as original equipment manufacturers (OEMs). Since they are in charge of the machine's design and manufacture, they can provide knowledgeable advice on installation, use, and maintenance.

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Industrial robot integrators provide robotic systems that are produced by original equipment manufacturers. Robot integrators, also referred to as "system" integrators, can provide a "turn-key" robotic work cell that is completely installed and includes guards, end effectors, and parts feeders. Robotic integrators can offer a wider selection of solutions since they can represent numerous firms and supply multiple products for each robot category.

Lesson-02: Advantages of Industrial Robots

This lesson focuses on the advantages of using industrial robots in a manufacturing process. Not only the higher initial outlay and costs, there are several tangible and intangible advantages. The efficient operation of industrial robots can pay for themselves in two to five years, making them a wise investment.

⚙️ **Faster Rate of Production:**

The main justification for investing in an industrial robot system is an increased production rate. Robots operate continuously without becoming tired or slowing down. If they are properly designed, operated, and maintained, they can effectively shorten manufacturing times. They process information far more quickly than humans do. This enables them to execute a rapid, deliberate sequence of motions, irrespective of intricacy.

⚙️ **Higher Load Capacity:**

Industrial robots utilize actuators that are significantly more powerful than those used by humans. The servo motor, often referred to as the robot's "muscle," plays a crucial role in this capacity. Modern robots can achieve impressive repeatability rates, sometimes exceeding 0.003 inches, thanks to these motors, which are precisely sized for the robot's requirements and capable of pinpoint accuracy. These servo motors can deliver substantial forces when coupled with highly precise planetary or cycloidal gearboxes. As a result, robots can effortlessly and efficiently move heavy loads that would pose significant challenges for human labor.

⚙️ **Improved Safety:**

Extreme temperatures, intense pressure, large loads, quick motions, and high-speed rotations are typical job risks. When doing procedures with these risks, industrial robots can help reduce the possibility of accidents or fatalities. Compared to their human counterparts, they are more resilient to hazardous working situations. Furthermore, since robots don't make mistakes or have bad judgment like human operators do, they increase workplace safety.

⚙️ **Lower Operating Cost:**

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Two of the financial advantages of industrial robots are reduced expenditures for human resources and less waste of raw materials. After investing in the robotic system, improvements in product quality, increased production rates, and operating savings are all attained. The efficient use and management of raw materials are made possible by the accurate and precise operation of robotic systems. This could lead to a decrease in the rates of product rejection. Manpower costs for manual labor operations are usually higher for the same quantity of effort. Numerous other expenses associated with physical labor, such as living allowances, government-mandated benefits, and training, are not related to the cost of operating a robotic system.

🌟 Better Repeatability and Precision:

Hundreds of thousands of cycles do not affect the behavior of a robotic system. The same set of duties can be carried out by robots reliably and consistently without the need for programming modifications or human participation. Its movement patterns, range of motion, force used, speed, and other operating characteristics are limited in their impact by external factors. Operating rates and product quality are hence consistent and predictable.

🌟 High Accuracy:

Robotic systems are built with greater operating precision than human operators. They can easily perform the exact actions that their application requires. This characteristic is essential for manufacturing processes that require exact tolerances, such as making auto and aircraft parts. Accuracy and repeatability in robotic systems are largely dependent on precision rotary encoders. They supply precise position data to the robot's controller, which it uses to govern the robot's movement profile, end-of-point location, and speed.

🌟 Excellent Product Quality:

Industrial robots are already a widespread fixture in many high-precision manufacturing processes. This is attributed to both reproducibility and accuracy. These characteristics allow the robot to produce products with dependable attributes free from defects caused by common errors and subjective evaluation.

When given the proper end effectors, robots may complete tasks quickly and effectively without sacrificing the caliber of the product. Even more advanced surgical procedures are performed by some robots under the supervision of medical professionals and the operating crew.

🌟 More Compact Production Area:

Because of their integrated end effector, larger load capabilities, and improved throughput, industrial robots can save space. It is not necessary to have additional equipment to help human operators perform their tasks. When employing human labor, a larger workspace is typically required to support more workstations and increase the manufacturing line's throughput. The same task might be completed by a single industrial robot.

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Lesson-03: Industrial Robot Applications

Industrial robots are most often used for simple pick-and-place operations. However, industrial robots are also employed in more significant and versatile activities because of advancements in control technology, robust actuators, and sophisticated sensors. The following are some of the most common applications for industrial robots.

Product Assembly:

Industrial robots are often used in assembly machinery. For accurate but highly repetitive tasks that would be taxing for a human operator, they function well. Its EOAT is usually mechanical grippers that quickly pick up, position, and orient tiny or large items. Optional sensors are typically used to recalibrate the robot's movement accuracy.

Modern end effectors usually feature sensors to ensure the proper part is present and orientated for the assembly process. Moreover, certain robotic systems can use a machine vision camera mounted to the end effector for dynamic positioning, accounting for assembly site changes that might happen during manufacture.

Non-Conventional Machining:

Abrasive jet, waterjet, laser, electric discharge machining (EDM), and plasma cutting are examples of common non-conventional machining techniques. Through the use of very focused streams of water, light, electric charge, or another physical entity, these non-contact machining technologies remove material. The substance is eroded, vaporized, or melted by the focused stream.

These procedures need a lot of energy, which, if not managed properly, could harm the finished item or the machine itself. Accurate control over the machine's cutting path is achieved through the usage of industrial robots. Digital industrial robots are used to maintain optimum cutting speed, stream stability, and precise control over machine parameters like power, pressure, and flow rate.

Palletizing and Depalletizing:

Palletizing is a technique used to handle, store, and distribute products more effectively by merging multiple distinct products into a single cargo. Conversely, depalletizing is the process of taking a palletized load apart. Due to their labor-intensive nature, both of these procedures run the risk of becoming bottlenecks. Robotic palletizers are employed due to their more economical and efficient product handling. Robotic palletizers incorporate EOATs, which are mechanical, pneumatic, or vacuum grippers that work by picking, orienting, and stacking objects in a manner akin to assembly line processes.

Welding:

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Robotic welding systems are widely used in numerous high-volume metal production processes and are frequently observed in automotive manufacturing plants. Higher operating rates and improved product quality became necessary as market competition grew. This calls for more precise and accurate welding procedures. Better control over many parameters, including current, voltage, arc length, filler feed rate, weld rate, and arc travel speed, is the primary benefit of utilizing industrial robots in welding.

Painting and Coating:

Painting and coating are delicate processes that call for precise movements that may be repeated to produce layers of consistent thickness. Painting calls for not just the necessary precision and accuracy but also the handling of potentially dangerous substances. Poisonous colors and solvents abound; some can even ignite an environment. Industrial robots are used to reduce all of these risks.

Grinding, Polishing, and Buffing:

Common secondary manufacturing techniques used to enhance the product's final look and surface characteristics include grinding, polishing, and buffing. The abrasive or polishing substance is moved in repeating, oscillating motions during these procedures. This is an easy movement for a robotic arm to emulate.

Deburring:

Deburring is another function of contemporary industrial robots. To smooth and deburr cast or injection-molded parts, a robot grasps a rotating tool—typically a wire wheel, sanding drum, or carbide deburr tool—and moves it along a predetermined path. The benefit of deburring with a robot is that the operator will typically come into contact with dust or debris during the process. Smoothing between pieces is more consistent when a robot follows a repeatable course.

Machine Loading and Unloading:

Robotic systems' high load capacity and mechanical advantage are utilized in machine loading and unloading. Transferring massive metal or plastic parts from casting, molding, and forging processes to conveyor systems, secondary processing stations, or filling machining centers with blanks for machining are some examples of specific machine loading and unloading uses.

Inspection:

To execute inspection activities on components or assemblies, robotic inspection systems can make use of measuring instruments such as optical sensors, proximity sensors, force transducers, ultrasonic probes, and even full machine vision systems. To ensure uniformity and quality, these

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devices are usually used to measure a product's dimensions exactly. Non-destructive testing (NDT) of welds is another application for inspections; in this case, ultrasonic probes or arrays are moved and controlled automatically by a robotic system.

Sorting:

Robotic systems' high-speed monitoring and straightforward pick-and-place capabilities are employed in sorting procedures. Visual sensors pick up on changes in shape, color, or size. A robot is employed to pick and reject items that are deemed unusual. Robotic sorting systems are frequently used in the electronics and pharmaceutical industries.

Lesson-04: Types of Industrial Robots

Industrial robots are categorized based on how their arms are set up. The parts of a robotic arm are joints and linkages. Changes in the quantity and kind of these two elements result in robots with various configurations. The few varieties of industrial robots are listed below.

✱ **Cartesian Robot:**

Three prismatic joints comprise a Cartesian robot. As a result, the tool can only produce circular motion along each axis when using kinematic models that support circular interpolation.

The three-dimensional Cartesian coordinate system, which comprises the X, Y, and Z axes, is the source of the term "Cartesian." The most basic type of robotic system is a cartesian robot, which can only move in a translational direction. They work well in applications where angular translations are not necessary and just right-angle movement is needed.

Unlike other robot kinds, cartesian robots can be designed to take higher loads because one or two of their prismatic joints can be supported at both ends. A gantry machine is an illustration of a Cartesian robot. The term "gantry machine" refers to a type of equipment used for picking and placing heavy, palletized items.

✱ **Polar Robot:**

Polar robots, sometimes known as spherical robots, operate using the three-dimensional polar coordinate system consisting of r , θ , and ϕ . Polar robots have a spherical range as opposed to a rectangular prism-shaped work environment. The radius of their range of motion is the same as the length of the link between the closest revolute joint and the EOAT. In comparison to other robot kinds, polar robots can reach the furthest for a given arm's length thanks to this arrangement. A prismatic joint connects a second link, which increases the range of a polar robot. Polar robots are frequently utilized in machine-loading applications due to their broad reach.

✱ **Cylindrical Robot:**

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A cylindrical robot can move in a cylindrical range, as the name implies. One revolute joint and two prismatic joints make up this type. The links can rotate around the robot's axis thanks to the revolute joint, which is situated at the base of the arm. The robot's cylindrical work envelope can be adjusted in height and radius using its two prismatic joints. In smaller versions, the prismatic joint that was previously used to adjust the arm's radius is eliminated. When performing straightforward pick and place operations and the product supply is only present in one location, this single revolute, single prismatic joint type comes in handy.

✱ **Selective Compliant Articulated Robot Arm (SCARA):**

A SCARA robot has an arm that is flexible or compliant in the XY plane, or horizontal direction, but rigid in the Z-axis, or vertical direction. Its "Selective Compliant" quality is characterized by its translational movement in a single plane. Two links, two revolute joints, and one prismatic junction make up a SCARA. The revolute joints, which are positioned along the same axis, attach the links and the base. The EOAT can only be raised or lowered via the prismatic joint. A SCARA produces a torus-shaped work envelope as a result. Its use is comparable to that of a round robot.

✱ **Delta Robot:**

A minimum of three links attached to an EOAT and a shared base make up a delta robot. Three undriven universal joints unite the EOAT to the linkages. Conversely, the base is joined by three joints that are powered either by a prism or a revolute. Together, the driven joints give the EOAT four degrees of freedom. In prismatic joint designs, rotation is typically facilitated by connecting a fourth link or shaft to the EOAT. A delta robot's EOAT can revolve around the vertical axis and move along all Cartesian axes to create a work envelope that resembles a dome. For high-speed pick and place tasks, delta robots are an excellent choice because of the three driven joints operating simultaneously.

✱ **Articulated Robot or Anthropomorphic Robot:**

The most popular type of robots utilized in manufacturing operations are articulated ones. They carry out more intricate tasks like machining, welding, and product assembly. Six degrees of freedom are intended for EOATs installed on articulated robots. There is a minimum of three rotating joints in the robot arm. To rotate the EOAT, a fourth revolute joint may be added to the arm's wrist. Like the polar robot type, it also has a spherical work envelope.

Lesson-05: Performance Specifications of Industrial Robots

When thinking about a particular robot, there are various design factors. Certain performance requirements for various applications must be balanced with the necessary expenditure. Improved specs are preferable, but the price rise is nearly exponential.

✓ **Axe count or degrees of freedom:**

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Robot mobility and EOAT orientation in three dimensions are determined by the number of axes and degrees of freedom. The types and directions of motion are taken into consideration when defining the degrees of freedom. The six directions of motion are yaw, pitch, roll, left or right, up or down, and forward or backward. A robot may have all six axes, depending on how it is designed; the more axes it has, the more freedom it has for its robotic arm.

✓ **Load Capacity:**

The weight or force a robot can apply to a load, as well as the amount of strain it can withstand, is measured as its load capacity. The weight capacity of robots ranges from 1.1 to over 2205 pounds, depending on the industry in which they are used (0.5 to 1000 kg). In pick-and-place applications, load capacity criteria are crucial.

Modern robots have stated load capacities that are dependent on their maximum payload, which considers the need to accelerate and decelerate the payload and end effector to and from the robot's maximum stated speeds. The robot that best fits the load capacity of an application is chosen by carefully weighing the end effectors that are integrated into it.

✓ **Accuracy and Repeatability:**

These are the two primary factors that determine how well a robot works to accomplish its tasks. The capacity of a robot to position itself or its load at a specific spot is known as accuracy, and it is determined by comparing the robot's final state to a user-defined set state.

The robot's capacity to hold its final position throughout multiple working cycles is measured by its repeatability. This is calculated by averaging the distances between the robot's point and the end positions of the EOAT following each cycle. While a pre-programmed position may always be "touched up" to bring the robot where it needs to be, repeatability is the most important factor to consider when choosing a robot because it determines the consistency of an operation.

✓ **Work Envelope:**

The swept area, stroke, and reach of a robot define its work envelope, which is its spatial specification. The area accessible to the robot's EOAT is a crucial component of the work envelope. The length of its links and the kind of robotic arm (Cartesian, polar, SCARA, etc.) determine how big the work envelope is. For applications involving machine loading and unloading, the work envelope's specifications are crucial. It must be decided before installation how much space is needed for a larger robot because it needs a larger work envelope.

To create safe zones for workers near robots, the work envelope serves the purpose of defining and controlling the region that a robotic arm may reach. Since everything a robot performs is related to its work envelope, it is imperative to understand the work envelope.

✓ **Maximum Speed and Acceleration:**

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Speed and acceleration have a direct impact on a robot's design operating speed and throughput. These characteristics are determined by the arm's kinematic design as well as the power ratings of the actuators, transmission, and driver in the robotic system. Even while speeding up a robot would seem like the best way to increase efficiency, slowing it down and speeding it up takes time, which reduces efficiency. Rapid alterations in velocity or significant acceleration might apply stresses to a robot that may jeopardize its quality, containment, or structure.

✓ **Controller Specifications:**

Each robot comes with a controller that may be used to control it and program its locations, speeds, and gripper clamps or unclamps. Robots from different brands are programmed using different techniques. Certain controllers allow for "teach through" programming, in which the robot is moved to different locations by using a "teach pendant," which is then used to "record" and utilize the position in the program. Making organized or sequenced code, which is akin to "C" or BASIC programming and involves creating a series of motions or I/O operations to accomplish a goal, is another technique that most controllers provide.

In industries, industrial robots are like super assistants. They can do tasks like welding and assembly quickly and efficiently thanks to their intelligent controls and powerful arms. Because they can employ sensors to gather information from their environment, these robots are intelligent. They collaborate with others without any issues, producing higher-quality work with fewer errors. Giving them instructions is one way to instruct them, and there are other approaches to doing so. Thus, industrial robots are the manufacturing equivalent of superheroes. They improve and streamline everything. These robots will play an increasingly bigger role in the way we manufacture things in the future as technology advances.

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Module Three: Industrial Automation Technologies

Industrial automation is the use of control technologies to reduce or eliminate manual labor in manufacturing. How is that accomplished? By incorporating computers and smart tools into several processes, industrial automation and controls contribute to system streamlining. Computers can operate manufacturing processes with little assistance from humans once they have been initially programmed.

Lesson-01: Industrial Automation Tools

Industrial control systems are one of the core technologies of industrial automation (ICS). Once more, industrial control systems encompass a broad range of control systems and associated software tools that are used to automate and manage a variety of industrial operations. Control systems come in a range of sizes and levels of sophistication, from entry-level controllers to sophisticated SCADA systems that can oversee industrial and manufacturing operations throughout several geographies and technological tiers. Let us examine some prevalent forms and constituents of industrial control systems.

- ✓ **Programmable Logic Controller (PLC):** A PLC is a digital industrial computer control system that has been ruggedized and is preprogrammed to perform automated tasks in industrial processes. To finish the task in the industrial process or machinery, the PLC continuously monitors and receives information from input devices or sensors, processes the information, and activates the connected output devices.
- ✓ **Supervisory Control and Data Acquisition (SCADA):** Industrial processes are controlled and observed using SCADA systems. Through direct communication with devices, such as sensors and PLCs, the system gathers and processes real-time data, logging events into a log file. SCADA is crucial for data analysis since it facilitates wise decision-making for industrial process improvement.
- ✓ **Human Machine Interface (HMI):** A human-machine interface, or HMI, is a software program that facilitates communication and interaction between a human operator and the machine, or production system. It allows for improved control over the production process and all of its applications by converting complex data into easily understood information.
- ✓ **Artificial Neural Network (ANN):** An ANN is a computational system that resembles a network of interconnected neuron nodes, much like the structure of the human brain. Artificial neural networks (ANNs) mimic how the human brain interprets and uses data.
- ✓ **Distributed Control System (DCS):** A DCS, or central monitoring network, is a network of connected devices used to control various components of an automated system.

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- ✓ **Robotics:** Robots can promote worker safety, enhance production flow and quality, and carry out jobs in hazardous or complex environments with efficiency. Furthermore, robots can greatly improve the convenience or comfort of daily living.

Conveyor belts were the beginning of industrial automation, but it has evolved significantly. Staying ahead of the competition means knowing what kind of automation to implement and how it will help the business. Managers will be better able to identify where the biggest gains are and which Industry 4.0 technologies can boost output, quality, revenue streams, and workforce as more data becomes available.

Lesson-02: On-premise vs. Cloud Systems for Industrial Automation

Cloud solutions and technologies are becoming more and more popular. Global end-user expenditure on public cloud services is expected to increase by 20.4 percent in 2023 to reach \$494.7 billion, up from \$410.9 billion in 2022, according to the most recent Gartner prediction. End-user spending is projected to approach \$800 billion by 2024.

Additionally, the number of manufacturers offering software systems for cloud-based industrial automation is expanding quickly. Given that cloud delivery models provide several advantages over on-premise solutions, it makes sense for enterprises to explore the prospect of making this transition.

Choosing between the two ways can be a challenging issue for decision-makers in firms and organizations aiming to introduce new automation technologies, as each has pros and cons. However, there may be long-term effects from this choice, and in most situations, there are, particularly for large firms with several processes and extensive operations.

On-premise vs. cloud systems: The deployment strategy, or, to put it another way, the location of the industrial automation software, is the sole essential distinction between cloud and on-premise systems. Nevertheless, this distinction is significant enough to affect an organization's automation strategy as a whole. Let's define on-premise and cloud in detail just to make sure we are all in agreement.

- ✓ On-premise industrial automation software is set up and kept up to date on the organization's local servers.
- ✓ Automation solution providers host and operate cloud-based industrial automation systems on their servers.

Naturally, depending on the particular demands and requirements of the company, one of these ways may be more advantageous than the other.

To ensure that you make the best decision, you must be fully aware of the advantages and disadvantages associated with each strategy. So, let's examine them in more detail.

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Lesson-03: Advantages & Disadvantages of On-Premise Automation Systems

Advantages of on-premise automation systems:

- ✓ **Full control:** The primary benefit of utilizing on-premise industrial automation components is that they are an internally produced solution that can be tailored and put into practice to meet certain demands and specifications. Additionally, the organization is fully in charge of the rest of its life cycle.
- ✓ **Better uptime:** As long as the organization's IT infrastructure meets this need, on-premise industrial automation systems may usually offer higher uptime and dependability because they don't depend on an Internet connection to work.

Disadvantages of on-premise automation systems:

- ✓ **Require large up-front investments:** Naturally, the biggest drawback of on-premise industrial automation systems is the large upfront costs associated with their creation and deployment.
- ✓ **Higher maintenance costs:** One such potential drawback of on-premise systems is maintenance costs. An on-premise system will steadily eat away at a company's budget as the expenses of maintaining any software solution tend to rise over time—maintenance charges make up 55% of the total cost of ownership of any software product.
- ✓ **More time to build and implement:** An on-premise automation system requires more time to create and implement. It takes time to customize the system and deploy it on servers and all devices. Potential issues, which are common for internal IT initiatives at businesses without software development experience, might cause delays and project budget overruns.

Lesson-04: Advantages & Disadvantages of Cloud Automation Systems

Advantages of cloud automation systems:

- ✓ **High accessibility:** One of the key benefits of all cloud solutions is remote accessibility from any device and location, and cloud industrial automation systems are no exception. Organizations can set up internal automation systems even if their IT equipment is dispersed across multiple locations and, in certain situations, continents by utilizing cloud technologies.
- ✓ **Lower costs:** The fact that cloud-delivered automation solutions and other SaaS products are far more affordable than on-premise hosting is another important advantage. By utilizing cloud solutions, businesses can avoid having to set aside money for the acquisition, setup, and upkeep of the on-site servers and other infrastructure needed to operate their industrial automation solution.

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- ✓ **Reliable performance:** If a third-party solution is offered by a respectable industrial automation provider, a business can outsource not just the maintenance of the automation system's physical infrastructure but also other hassles associated with the on-premise strategy. which can be sufficient given that modern industrial automation systems must operate continuously, deliver high and steady performance, be interoperable with an expanding range of software solutions and platforms, receive regular and timely updates, etc.
- ✓ **Financial predictability:** Financially speaking, cloud-based industrial automation systems can be quite advantageous, not only because they are typically far less expensive to use and have no upfront expenditures. The fact that the prices are far more predictable is another important consideration. This makes it possible for enterprises to plan for automation and eliminate concerns about unforeseen costs.
- ✓ **Flexible and scalable costs:** Furthermore, most cloud process automation technologies offer customizable subscription plans according to an organization's present needs and specifications. Because the expenses of automated support will only rise with more business activities, this helps organizations expand operations more easily and avoid overspending on IT infrastructure.
- ✓ **Better security:** In terms of safeguarding industrial automation systems against diverse cyber threats, such as hacker attacks and data breaches, cloud-based solutions often provide a more robust security framework. The majority of small and medium-sized businesses cannot afford the strong centralized cybersecurity systems used by the large data centers where cloud providers often house their servers.
- ✓ **Quick deployment:** Lastly, cloud solutions can be deployed considerably more quickly and easily than on-premise systems. Instead of taking months or more, enterprises may deploy industrial automation technologies in a matter of days or weeks.

Disadvantages of cloud automation systems:

- ✓ **Lower customization potential:** On the other side, a business utilizing a cloud industrial automation system would normally have less control and fewer customization possibilities because it is a third-party solution.
- ✓ **Dependence on the solution provider:** While using a cloud system rather than building an autonomous industrial automation solution enables businesses to outsource the majority of the challenges associated with its creation and upkeep, it also highlights the significance of the third-party supplier. Because of this, when evaluating their automation alternatives, factory automation integrators, IT managers, and business decision-makers also have to choose the proper provider.
- ✓ **Internet connectivity required:** For cloud automation solutions to function effectively, they require Internet access. This can undoubtedly be an issue in certain situations, particularly for businesses that need total independence from the Internet from their industrial automation systems.

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In light of everything discussed above, industrial automation solutions, whether on-premises or in the cloud, are still useful and important. However, cloud systems—a component of Industry 4.0 technological advancements—are undoubtedly becoming more and more well-liked since they significantly reduce the cost and increase the accessibility of automation technologies, even for small and medium-sized businesses that lack the resources to support an on-premise solution.

Technologies for industrial automation represent a major advancement in the intelligent and productive work environment. Together, these technologies—which include sensors, robots, and smart systems—help factories and other businesses run more smoothly. They increase accuracy and consistency in addition to increasing production by eliminating manual labor and automating repetitive operations. The foundation of smart manufacturing is the integration of several technologies, which opens the door for Industry 4.0. Industrial automation has a noticeable effect on both production speed and product quality as facilities become more responsive and networked. Essentially, these technologies are reshaping industries to become more flexible and responsive to the changing needs of the modern world, paving the way for a future in which machines work together effortlessly.

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Module Four: Robot Programming and Integration

To fully realize the potential of industrial robots, programming and integration are essential. Robots require exact instructions to function properly because they are complex devices that can do intricate jobs. Creating these instructions and directing robots through a variety of motions and behaviors is known as robot programming. The goal of integration, on the other hand, is to smoothly integrate robots into already-existing systems to produce a streamlined and effective workflow. To fully utilize automation, whether for assembly, manufacturing, or other uses, robot programming and integration must be mastered. By enabling companies to fully utilize robot capabilities, this dynamic field ushers in a new era of production and precision.

Lesson-01: Robot Programming Basics

You can use any programming approach, but there is a standard procedure that is the same for all of them. This is a dependable method for programming robots, regardless of your level of knowledge, from novice to expert.

- ✓ **Pick the Right Programming Platform:** Selecting the programming method is the initial step in any project. Which platform and software are you going to use? You may choose to hard code the robot's programming using the manufacturer's programming language because you have extensive familiarity with it. As an alternative, you might go with an offline programming program that is easier to use.
- ✓ **Plan Your Program:** When learning to program a robot, a common novice error is to begin the coding process right away. Seldom does it work out nicely.
Plan your robot program well before implementing any programming technique. Choose the actions that the robot must perform. Using your hand and arm to "play out" the program is a quick and easy method to accomplish this.
- ✓ **Create the Program:** It's time to start writing code! The specifics of this phase will vary based on the programming approach you select:
 - First, initially, training the robot using its teach pendant involves manually positioning it through a process called 'jogging' and recording each point into the software.
 - Second, typing out the commands and continuously testing them on the actual robot is the programming process for text-based programming in the language of the manufacturer.
 - Lastly, programming with graphical offline programming only entails utilizing point-and-click commands to move the virtual robot.
- ✓ **Do Copious Debugging:** When programming a robot, debugging is essential regardless of the programming technique you choose. This entails putting the software to the test on the actual robot using traditional methods of robot programming. This frequently translates into extended periods of inactivity for the robot, which lowers its output.

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- ✓ **Put the Robot into Production:** Putting the robot into production and seeing a return on investment is the last step in any robot programming project! This includes cleaning up the robot cell, connecting it to other operations, and testing the robot's software one last time.
- ✓ **Collect data and improve:** You shouldn't "set it and forget it" when it comes to your robot cell, even when your programming assignment is done. Gather robot operating data so you can monitor its progress over time. Evaluate your data once in a while to see how well it is working. You'll probably find ways to fine-tune the robot's programming to enhance its functionality even further. You can test your modifications on the simulated robot without affecting its productivity if you opt for offline programming.

Lesson-02: Programming Languages for Industrial Robots

There are several programming languages available, and each has benefits of its own. Since Java is the most widely used programming language, your coworkers will probably be most able to assist you with it. But in the robotics business, Python and C are almost equally popular and well-liked programming languages. A wide range of engineers may now easily grasp and use programming thanks to visual programming languages like LabVIEW. Programming languages vary in their benefits, depending on your level of experience. So, which one fits you the best?

The Advanced Programmer:

You are likely familiar with Java, C, and Python if you have prior experience with robotics development and coding. Specifically, real-time high-level programming and embedded development are done with the C language. This covers algorithms for motion decision-making, control, and visual processing. These are the fastest programming languages, supporting RAI, multi-threading, object-oriented programming, and memory management the best.

Novice programmer:

If you have some programming experience but still require some guidance and support, we recommend MATLAB, LabVIEW, or Java. The internet is replete with materials for all three of these widely used, tried-and-true languages. You likely have experience with Java and MATLAB if you majored in engineering in college. LabVIEW offers a welcome change of pace. National Instruments' proprietary software is this visual programming language. It is integrated into myRIO and the old cRIO products. Should you have taken part in FIRST Robotics or Vex in high school, you might also be acquainted with LabVIEW. Even while LabVIEW is an excellent tool for novices, experienced programmers can also benefit from its capability.

Programming Newbie:

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Python, Arduino, and visual programming languages like LabVIEW are ideal if you are familiar with programming concepts, or if you would just like to brush up on them. An essential language for programs like microcontrollers, Python is simple to learn. Nevertheless, real-time vision processing, iterative procedures, and applications needing precise DOF control are not appropriate for Python. What you receive in online libraries and assistance is the trade-off. Experts in Python provide some of the most comprehensive and detailed materials. For offline programming, this is the best programming language.

Bonus Languages for AI:

Regardless of your degree of experience, LISP would be a fantastic choice for you if you plan to work on AI and programming a lot. List structure is used by LISP to represent data. Because data and programs have the same structure, LISP programs can function as data on other systems. This indicates that LISP programs are perfect for "learning" self-modifying programs, or artificial intelligence. LISP is a fantastic auxiliary language that any roboticist should be familiar with. It is internally reflexive, robust, and easy.

Lesson-03: Integration of Robots into Existing Workflows

The efficiency and precision of current workflows are revolutionized by the integration of robots. Businesses may improve productivity, minimize errors, and optimize processes by integrating automation with human operations in a seamless manner. Robots are great at time-consuming, repetitive jobs, freeing up employees to concentrate on higher-value tasks that call for creativity and analytical thought. This integration results in a dynamic and responsive workflow that not only speeds up production but also guarantees smooth man-machine collaboration. As industries embrace this synergy, they open up new opportunities for technical innovation, cost savings, and growth in the dynamic world of modern workplaces.

- ✓ **Assess your needs and goals:** Establishing your goals and expectations, as well as your actual opportunities and obstacles, is the first step towards integrating robotics. What obstacles and snags exist in your production process? Which abilities and jobs are demanding, monotonous, or dangerous for your employees? What are the advantages and disadvantages of automation for your company? How do you assess and keep track of your output and caliber? You can ascertain the size, scope, and viability of your robotics project by responding to these queries.
- ✓ **Choose the right robots and tools:** The next stage is to choose the kind and model of tools and robots that best meet your objectives. Different types of robots, with varying characteristics, capabilities, and needs, exist. These include industrial, collaborative, mobility, and service robots. The software, sensors, controls, grippers, and other add-ons that let the robots do their jobs must also be considered. It is advisable to assess the robots and tools' technical parameters, compatibility, dependability, safety, and accessibility to maintenance and support.

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- ✓ **Design your layout and workflow:** Creating an architecture and workflow that can fit the robots and optimize your process is the next phase. You must consider your facility's physical layout, infrastructure, and space requirements as well as how the robots will interact with the equipment, supplies, and personnel. The parameters, steps, and sequences of your process, as well as how the robots will carry them out, must also be mapped out. A workflow that minimizes waste, errors, and downtime should be fluid, adaptable, and efficient.
- ✓ **Implement and test your solution:** Implementing and testing your solution to make sure it functions as planned is the fourth phase. The robots and tools must be installed, programmed, and integrated into your system while adhering to safety and quality requirements and best practices. Additionally, you must find and fix any problems or gaps in your solution by testing and validating it in a variety of circumstances. Involve your employees and stakeholders in this process, and make sure they receive the necessary assistance and training.
- ✓ **Monitor and improve your performance:** The last phase is to track and enhance your performance to get the most out of your robotics system. To assess your outcomes and feedback, you must gather and examine data on your workflow, product, and quality. Additionally, you must constantly check on, maintain, and upgrade your tools and robots. In addition to staying current with the newest trends and advancements in robotics, you should look for chances for innovation, adaptation, and improvement.

Robotics' incorporation into current processes is a revolutionary step toward a future that is more productive, inventive, and collaborative. This combination of robotic accuracy and human knowledge unlocks previously unheard-of productivity, simplifies processes and creates a harmonious synergy between automation and human creativity. Industries that accept and adjust to this technological innovation not only increase productivity but also create new opportunities. Together, robots and humans are redefining the workplace and ushering in a dynamic era when human creativity and robotic efficiency combine to propel us to previously unheard-of levels of productivity and advancement.

Lesson-04: The Benefits of Robotics Integration in Manufacturing Industry

Robots in manufacturing provide numerous benefits to the company since they do various activities more efficiently than human workers and remove the drudgery from many operations. Robots can operate continuously, execute tasks with great accuracy, and produce consistently

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good outcomes. Due to the rapid payback period, this results in much higher productivity and better-quality output at a lower cost. Other benefits of robots include improved floor space use, heightened safety, and operational flexibility. A few of these advantages are covered in detail in the paragraphs that follow.

- ✓ **Enhanced Productivity and Operational Efficiencies:** Robots are faster, more accurate, and more consistent than human workers at any given task, which improves operational efficiency and production. The same amount of work is completed significantly more quickly and efficiently, without any pauses or interruptions. This is one of the main benefits of using robots since a survey found that human error accounts for over 25% of unexpected downtime in the manufacturing industry, compared to roughly 10% in other industries. Here, it's critical to keep in mind that robotic automation boosts efficiency even more if it already increases it.
- ✓ **Improved Quality Consistency:** Robots, in contrast to human laborers, do not experience fatigue from repetitive tasks or attention deficit disorder, which would lead to subpar work. After being trained for a particular activity, the robot operates flawlessly and consistently, producing welds of the same quality from Job 1 to Job 1000. This also applies to robots that tend to machinery, which labor continually to load and unload tasks into them. This produces work that is not only of excellent quality but is also continuously of that quality.
- ✓ **Increased Workforce Safety:** Regardless of how diligent a worker is or how well-maintained a facility is, there is always a chance of an incident or accident brought on by a slip-up or human error. In addition, a lot of places of employment, such as paint and welding shops, foundries, and heat treatment centers, have dangerously high temperatures and poisonous gasses that might harm people's health. In such conditions, robots can function flawlessly and maintain their normal efficiency. Robots follow preprogrammed movements exactly after they are trained to do so. For added safety, they are also working in confined spaces.
- ✓ **Better Factory Space Utilization:** Human workers require space to feel comfortable and avoid claustrophobia. All living things require space to allow for freedom of movement, and humans are no different. Robots have a tiny footprint because they don't require additional space beyond what they can move, which improves shop floor usage. Compared to an assembly line with human workers, an automated robotic one will require a lot less area.
- ✓ **Escalating Throughput and Profit Margins:** A study comparing the productivity of a manufacturing plant with only human labor and one using industrial robotic automation found that the use of robots increased productivity by 50%, which in turn led to a corresponding 50% increase in productivity and a more than 85% increase in facility utilization. Higher profit margins and higher throughput are the immediate results of this. Further benefits could result from having a production plant with robotic automation run

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around the clock, as many facilities in more developed nations have done for years, with a small number of human supervisors working the night shift.

- ✓ **Scalability and Flexibility:** Robots have both scalability and flexibility. A manufacturing plant can increase production by adding more robots after they are integrated into its processes. In contrast to human laborers who possess specialized talents, robots may be programmed to perform a wide range of tasks. This adaptability allows them to be used in a very flexible manner. In the era of mass customization that Industry 4.0 envisions, this flexibility is particularly desirable.

No manufacturing facility can afford to overlook the fact that robots may increase productivity significantly and generate higher profits thanks to their many benefits. Some businesses must use robots extensively if they want to compete in the global manufacturing market. But it's also crucial to first research the precise specifications and, more importantly, whether or not robots are genuinely appropriate for a particular industry or operation. If a plant can function financially without robots, then spending a lot of money on robotics could be detrimental if the return on investment is not adequately planned and there is no matching gain in production or efficiency. This is where professional guidance is most helpful in weighing the benefits and drawbacks of robotic automation.

The foundations for realizing the full potential of industrial automation are robotic programming and integration. Learning how to program robots to follow exact instructions guarantees their effectiveness and flexibility. The smooth integration of these devices with current systems improves productivity by streamlining processes. The need for skilled robot programming and integration is growing as industries continue to move toward further automation. This mutually beneficial partnership between industry and technology portends a world in which robots function with greater ease and contribute to an environment that is more accurate, responsive, and efficient. Essentially, if one wants to navigate the revolutionary journey towards advanced and interconnected industrial processes, mastery of robot integration and programming is essential.

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Module Five: Safety Standards and Regulations

Industrial robots require careful attention while being used in the workplace. When working with robots and people together, it's critical to recognize any dangers, reduce them, adhere to safety regulations, and uphold best practices and safety standards at all times.

The primary cornerstones of safeguarding people's health and the integrity of operations in a variety of businesses are safety standards and regulations. The significance of establishing a safe environment for employees and the general public is emphasized by these principles, which provide a framework for identifying, evaluating, and mitigating possible risks in the workplace. Respecting safety regulations not only saves lives but also encourages dependable, regular procedures. These rules essentially act as a cornerstone for promoting responsible behavior, creating a culture of safety, and reducing risks in various professional contexts.

Lesson-01: Industrial Robotic Safety Risk Assessment

Although industrial robots provide numerous financial advantages for businesses, they are also intrinsically dangerous, and businesses must provide a safe working environment for their employees. Although many people consider safety to be a hassle, industrial robotic safety can offer numerous advantages, such as greater factory floor efficiency and the guarantee of regulatory compliance. Industrial robot safety concerns must be evaluated, but it can be challenging to know where to begin.

What's involved in an Industrial Robotic Safety Risk Assessment?

The first step towards attaining robotic safety in a production setting is evaluating the dangers associated with industrial robotic technology. A thorough risk assessment for industrial robots usually consists of two parts: an analysis and an evaluation of the risks.

Determining the robotic machinery's limits, identifying potential production hazards, and assessing the risk of detected hazards are the three processes that make up a risk analysis, which can be the more complex process of the two.

Although these precise stages may not be feasible for every firm, they serve as a useful framework for conducting an accurate industrial robotic risk assessment.

Identifying Sources of Hazard in Industrial Robotics: In a safety risk assessment, identifying hazards in industrial robotic operations could be one of the most crucial phases. Risks may manifest in material, electrical, thermal, noise, vibration, radiation, or mechanical forms.

Your application and the circumstances in which industrial robots are engaged—during integration, training, maintenance, or regular production, for example—will determine the kind of hazard that arises. The types of dangers and interaction situations must be evaluated in light of the safety procedures implemented to guard against worker harm or injury.

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Finding the sources of danger in industrial robots is an important but sometimes difficult task. Evaluating safety concerns is made easier once the risks are recognized and the robotic interaction cases are confirmed.

In industrial robotics, safety is critical. Manufacturers can even achieve regulatory compliance and better productivity in the process of enhancing safety. Manufacturers have a responsibility to protect workers from damage or injury.

Lesson-02: Safety Advantages Industrial Robotics Offer

AI automation combined with industrial robots has become a crucial component of the manufacturing process that creates the majority of goods in use today. Industrial robots can perform activities that humans would typically perform in the event of an explosion, infection, or other dangerous circumstances. Robotics operators or designers can equip robotic arms with specially made grippers to lift large loads and hold delicate objects at the same time. Robotic arms may be outfitted with almost any tool to guarantee exceptional speed, accuracy, and precision that greatly exceeds human capabilities.

To make viewing images easier, robotic arms can also be fitted with various sensors and visual systems. For the robotic arms to behave logically, the vision and sensor feedback can cooperate with AI.

- ✓ **Protection from Environmental Hazards:** Industrial robots increase worker safety by addressing common workplace environmental hazards. Robots can handle situations where there are significant risks for workers, such as severe noise, harsh temperatures, high dust levels, or exposure to chemicals. Safety managers can better concentrate on value-added duties by using industrial robots instead of spending time on simple safety problems like wearing safety glasses and earplugs.
- ✓ **Protection from Mechanical Hazards:** Humans are protected from various hazards such as being struck by, caught in, or caught between by industrial robots that carry out jobs. These risks are not totally removed because the robots themselves may still provide these particular risks (more on that below). There is a lot of debate over whether robotics offer greater protection than risk. However, indeed, robots often protect against mechanical risks. Creating a "no-touch" zone during operations significantly increases worker safety and reduces risk.
- ✓ **Improving General Safety Awareness:** Using industrial robots also has the benefit of raising awareness of workplace safety. Workers may lose focus after prolonged lengths of time spent on monotonous chores. It makes sense that employees might become sidetracked when constantly tightening the same bolts on a manufacturing line, for instance.
Additionally, accidents tend to happen to personnel who are disengaged. However, workers can handle fewer repetitive duties because robots can put the same bolts for

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eight hours straight. Because of this automation, employees are more aware of their surroundings, which improves the facility's safety culture.

- ✓ **Minimizing the Potential Risk of Repetitive Injuries:** Industrial robots increase awareness of potential risks that workers may encounter from ergonomic concerns, just as they do for ordinary environmental risks. Even something as simple as screw installation can include dangers such as carpal tunnel syndrome and other physical ailments from repetitive motion.

Lesson-03: Safety Standards Used in Robotics

Although there are no particular safety requirements for industrial robotics, the Occupational Safety and Health Administration (OSHA) provides safety concerns. OSHA does provide relevant General Industry standards, such as 1910.147 for lockout/tagout control of hazardous energy, 1910 Component O for equipment and machine guarding, and 1910 Subpart O for the selection and application of work practices for the Electrical subpart (1910.333).

OSHA further states that norms from the American Welding Society (AWS), the American National Standards Institute (ANSI), and the International Organization for Standardization (ISO) are probably applicable to contemporary industrial robots.

Manufacturing sectors adhere to ANSI safety regulations while using industrial robots. ANSI specifies standards for installation, remanufacturing, manufacturing, and safety procedures to protect people working with robots.

In ANSI/RIA R15.06-2012, ANSI provides safety guidelines for industrial robots and robot systems. These standards address safety design as well as potential risks related to robot hazards and procedures for mitigating or eliminating those risks.

ANSI and the Robotics Industries Association (RIA) have released safety guidelines for collaborative robots in RIA TR R15.606-2016 as an addition to the standards mentioned above.

Additionally, the ISO outlines safety requirements for industrial robots. For instance, the standards in ISO 10218-1:2011 are comparable to those in the first ANSI paper (ANSI/RIA R15.06-2012) that was previously cited. Being the only representative of the United States and one of the founders of ISO, ANSI aims to support and collaborate with ISO.

Apart from endorsing other organizations' standards, OSHA has collaborated with the National Institute for Occupational Safety and Health (NIOSH) and the Association for Advancing Automation (previously known as the Robotic Industries Association) to improve understanding, investigation, and dissemination of industrial robotics safety. OSHA announced revisions to the OSHA Technical Manual on Industrial Robot Systems and Industrial Robot System Safety in January 2022 as a result of this partnership.

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Lesson-04: Hazards When Working with Robots

Numerous possible hazards are present in the context of industrial robot safety and operation. The following is a summary of the main risks that arise when interacting with robots:

- ✓ **Installation Hazard:** Bad installation is one of the key risks associated with using robots in the workplace. The workplace can be unsafe due to improper robotic system installation and disregard for testing procedures, which provide potential risks.
- ✓ **Power System Hazard:** The risk of being exposed to electrical currents increases with the amount of human interaction with robots. By controlling hazardous energy (lockout/tagout), industrial robot dangers can be reduced and unauthorized access can be prevented. Robots continue to pose a risk to operators and other people in the vicinity as long as they are powered on or connected to power sources.
- ✓ **Controlling Hazard:** Robot movements can become inadvertent, inaccurate, and ineffective due to control mistakes including electromagnetic interference and software problems, which might potentially jeopardize the control system as a whole.
- ✓ **Human Error:** Human mistake is frequently eliminated from labor by robots. However, if there was any human fault in the robotic arrangement, it would indirectly affect the work being done. Robotic mishaps are frequently caused by mistakes like redundant programming, inadequate setup, and poor maintenance.
- ✓ **Mechanical Hazard:** The unexpected nature of this kind of potential hazard increases worker danger on the job. The dangers posed by moving machine parts can be decreased using machine guarding. However, there may still be a chance of being struck by, stuck in between, and other comparable concerns when using robotics.
- ✓ **Environmental Hazard:** This kind of possible risk results from outside variables that the employees cannot control. In any event, this affects a robot system's ability to function by causing a power outage or cable problems.
- ✓ **Access Hazard:** Unauthorized workers frequently lack knowledge about the safety requirements to enter the restricted area and operate the robots. Workers who are not properly trained or informed are less aware of "how" to work with robots, which frequently results in serious events.

How Companies Reduce Incidents with Industrial Robotics: Make sure that your occupational health and safety (OSH) management plan considers robots if they are a part of your workplace. To determine the degree of risk that industrial robots pose at your specific job site, conduct a risk assessment.

Adhering to safety regulations and giving priority to safety training are the best approaches to minimize mishaps involving industrial robots and safety. Since automated robot systems are now used by the majority of businesses, it is essential to protect employee safety. Safety managers should, for example, begin by labeling workspaces clearly.

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In the same vein, personnel should be trained and informed by safety coordinators about floor layouts and how to work around robots. Safety managers' primary goal in industrial robot safety should be to adhere to the most recent ANSI and OSHA robotics standards as well as applicable OSHA regulations. Workers who are physically near robots at work should be properly protected and trained.

Due to the widespread recognition of the critical role that safety plays in the deployment of industrial robots, strict safety regulations and cutting-edge technologies have been developed to reduce workplace accidents. The need to ensure the safe operation of industrial robots has grown as these machines become more complex and are integrated into a wider range of industries. Safety-focused programming and sensor-equipped collaborative robots, or cobots, represent advances in safer human-robot interactions. To further reduce risks, thorough safety protocols, routine maintenance, and worker training programs should be put into place.

As a result, these actions have greatly decreased workplace mishaps, promoting a safer and more efficient atmosphere. The industry's dedication to safeguarding human workers while maximizing the productivity of robots is shown in the ongoing advancement of safety technology and procedures, which will shape the future of industrial robotic safety.

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Module Six: Future Of Industrial Robotics and Automation & Case Studies

Lesson-01: The Role of Robotics in The Fourth Industrial Revolution

Industry 4.0: Understanding "the fourth transformation," also referred to as "industry 4.0," is essential. The human race has seen three industrial revolutions since the 1800s, each characterized by the arrival of innovative new technology that improved production and procedures. The production line, the computer, and the steam engine were the catalysts for earlier innovations. The term "Industry 4.0" refers to the ongoing industrial revolution that is being fueled by developments in robotics, automation, and the Internet of Things (IoT). The phrase "industry4.0," coined in 2011, heralds a new era of digital integration and smart systems that is permeating practically every industry on the planet.

The Internet of Things (IoT), artificial intelligence, and digital technologies are being incorporated into many facets of industry and society, marking the beginning of the fourth industrial revolution (4IR). A key player in this revolution, robotics is changing manufacturing, healthcare, transportation, and a host of other industries. The following are some salient features of robotics's place in the fourth industrial revolution:

- ✓ **Automation and Efficiency:** Automation of several jobs is made possible by robotics, which boosts productivity, efficiency, and cost-effectiveness. Modern robotic systems are capable of accurately and quickly completing complicated and repetitive tasks, which eliminates the need for human participation in some operations.
- ✓ **Smart Manufacturing:** A key element of smart manufacturing systems is robotics, which allows machines to talk to one another and make choices in real-time using data. Cobots, or collaborative robots, assist human workers in the production process, increasing efficiency and adaptability.
- ✓ **Internet of Things (IoT) Integration:** IoT sensors and devices are frequently combined with robotics to create networked systems that have data collection and sharing capabilities. Robots can perform at their best, adjust to changing circumstances, and add to the larger network of interconnected devices thanks to this integration.
- ✓ **Artificial Intelligence (AI) and Machine Learning:** Robotics improves decision-making and problem-solving skills by utilizing AI and machine learning techniques. Robots are capable of adapting to new environments, learning from past experiences, and continuously enhancing their performance through data analysis.
- ✓ **Human-Robot Collaboration:** Because they are made to securely operate alongside people, collaborative robots promote a more dynamic and adaptable workplace. The combined strengths of humans and robots are enhanced by this partnership, improving overall quality and productivity.

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- ✓ **Autonomous Vehicles and Drones:** Transportation and logistics are being revolutionized by autonomous cars and drones, which are being developed mostly due to robotics. In a variety of sectors, including logistics and delivery, autonomous systems can boost productivity, lower transportation costs, and increase safety.
- ✓ **Healthcare and Assistive Robotics:** With the development of surgical robots, robotic exoskeletons, and assistive technology for the old and crippled, robotics is significantly improving healthcare. These technologies help with rehabilitation, increase the accuracy of medical treatments, and raise the standard of living for people who struggle with movement.
- ✓ **Supply Chain and Logistics:** By using autonomous robots for order fulfillment, inventory management, and warehouse automation, robotics is significantly improving supply chain and logistics operations.
- ✓ **Customization and Flexibility:** More and more robotic systems are being made to be flexible and adaptive so they can be quickly reconfigured and reprogrammed to satisfy shifting production needs.

The fourth industrial revolution is centered on robotics, which propels innovation, efficiency, and connection across a range of industries. The way we work, create, and engage with the world around us is changing as a result of robotics' integration with other technologies like AI and IoT.

Lesson-02: Future Trends in Industrial Robotics and Automation

Industrial robotics is developing at an exponential rate; new discoveries are made every year that could fundamentally alter the way we manufacture, run, and automate processes. Numerous important technological advancements are causing a substantial shift in industrial robotics and automation. These patterns mirror continuous advancements that augment the capacities, adaptability, and amalgamation of robotic systems across diverse sectors. These are a few noteworthy upcoming trends:

- **AI-Powered Robotics:** Industrial robots are increasingly being controlled by AI. Thanks to machine learning algorithms, robots that were previously considered to be only seen in science fiction can now do tasks like making judgments and adapting to changing conditions.
- **Collaborative Robots (Cobots):** Cobot usage is increasing, and workplace integration of these machines is expected to increase by 2024. These robots offer greater flexibility and safety while living in harmony with humans.
- **Advanced Sensing Technologies:** Robots are becoming more perceptive thanks to enhanced sensors, such as force/torque sensors and 3D vision systems, which let them negotiate challenging surroundings and communicate more naturally.
- **IoT and Industry 4.0 Integration:** Manufacturing processes are being optimized through seamless communication, real-time data analysis, and predictive maintenance that result

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from the integration of industrial robots with Industry 4.0 and the Internet of Things (IoT) platforms.

- **Edge Computing:** Robots now can digest data instantly thanks to edge computing, which lowers latency and enables fast, intelligent decision-making in dynamic contexts.
- **Robotic Process Automation (RPA):** Automation of repetitive operations with robots is becoming more and more common, especially in manufacturing and logistics, as it increases productivity and lowers mistake rates.
- **Customization and Flexibility:** The flexibility and customization of industrial robots are increasing, enabling them to be tailored to a wide range of industry-specific requirements and procedures.
- **AI-driven Quality Control:** Robots with AI capabilities are currently leading the way in defect identification and quality control, guaranteeing accuracy and consistency in production processes.
- **Human-Machine Collaboration:** Humans and robots will work together even more closely in 2024 as AI makes robots more sensitive and perceptive, increasing both productivity and safety.
- **Energy Efficiency:** One of the industry's main forces is sustainability. The environmental impact of industrial automation is being lessened with the help of innovations in sustainable solutions and energy-efficient design.

Together, these developments show how industrial robots and automation are continuing to advance, with an emphasis on improving overall system performance, intelligence, adaptability, and safety. These advancements are probably going to have an impact on how industrial robots develop in the future.

Lesson-03: Real-World Incidents Involving Industrial Robotics

The risks that industrial robotics can pose are also demonstrated via incident examples.

An assembly line at an Alabama auto parts factory suddenly resumed when a worker went into a robotic station to do maintenance. Two weeks before her wedding day, the robot crushed her to death. OSHA issued citations and suggested fines totaling \$2.5 million for several infractions, such as disregarding energy control protocols, exposing employees to risks without protecting stored energy, and neglecting to install appropriate machine guarding.

An employee at an Ohio-based firm came into contact with a machine component on a robotic press line. Due to his arm's crushing, amputation surgery was necessary. Investigators found that concerning the machine's danger zone, the corporation did not provide enough protection. OSHA has recommended a total fine of \$3.42 million for this manufacturer due to a combination of offenses.

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In Washington, there was another incident at a company that bottled water. A warehouse dock coordinator was struck by a laser-guided vehicle (LGV), an autonomous forklift, resulting in fatal injuries. Because the worker was out of the sensor's line of sight, investigators believe the sensor was intended to trigger a stop but failed to detect it. There had been no emergency stop initiated to avert such catastrophes.

Lesson-04: Case Studies in Various Industries

Case Study-01: Tesla's Implementation of Industrial Robotics and Automation

As a pioneer in the automotive sector, Tesla never stops pushing the envelope on innovation. This case study explores how Tesla strategically integrated automation and industrial robotic systems to transform its manufacturing processes, emphasizing the company's dedication to efficiency, sustainability, and cutting-edge technology.

Background:

Leading the way in electric cars and renewable energy, Tesla had to balance growing demand with its dedication to premium, environmentally responsible manufacturing. Tesla started a revolutionary path to incorporate industrial robotics and automation into its manufacturing facilities to address these issues.

Objectives:

- ✓ **Sustainable Production:** Adopt environmentally friendly production techniques to support Tesla's sustainability objectives.
- ✓ **Efficiency and Scalability:** Increase production efficiency to keep up with demand and expand your business.
- ✓ **Innovation:** Adopt cutting-edge technologies to gain a commercial advantage in the automobile industry.
- ✓ **Quality Assurance:** Ensure that production standards are high-quality and consistent.

Implementation:

Tesla implemented a complete solution by utilizing its in-house expertise and working with top robotics and automation specialists.

- ✓ **Robotic Assembly Lines:** used robotic arms to precisely assemble electric vehicle parts and automated difficult operations like assembling the powertrain and installing batteries.
- ✓ **Automated Quality Control:** Applied machine learning techniques to continuously improve flaw detection through the use of artificial intelligence and sophisticated visual systems for real-time quality inspection.

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- ✓ **Material Handling Automation:** implemented robotically sorted conveyor systems for effective logistics and automated guided vehicles (AGVs) for smooth material transportation inside large plants.
- ✓ **Human-Robot Collaboration (HRC):** Collaboration robots were introduced to support human technicians and better safety precautions with real-time monitoring and cutting-edge sensors.

Results:

- ✓ **Sustainable Production Practices:** achieved notable decreases in waste and emissions and lessened environmental impact by using sustainable materials and energy-efficient robotic technologies.
- ✓ **Efficient and Scalable Production:** increased production capacity by 40%, exceeding and meeting market demand; additionally, scalability was made possible by quick adjustments made in response to shifts in production needs.
- ✓ **Innovation and Technological Edge:** By implementing cutting-edge robotics and automation and utilizing cutting-edge technologies to create novel production processes, Tesla was able to establish itself as a leader in the industry.
- ✓ **Quality Assurance:** Attained a previously unheard-of degree of assembly precision, guaranteeing constant high standards, and minimized flaws by using automated quality control procedures.

Challenges and Lessons Learned:

- ✓ **Continuous Innovation:** navigating the difficulties of keeping up with technology developments and placing a strong emphasis on continuing research and development to keep a competitive edge.
- ✓ **Integration and Training:** ensuring the smooth integration of robots into current procedures and giving staff members intensive training on how to use and maintain the new equipment.
- ✓ **Public Perception and Communication:** transparently communicating with stakeholders to address concerns about job displacement and to highlight the benefits of high-tech job growth and skill development.

In addition to revolutionizing its production procedures, Tesla's adept integration of automation and industrial robotic systems has raised the bar for sustainability, productivity, and creativity in the automotive sector. This case study demonstrates Tesla's dedication to expanding the realm of possibility and reaffirms the company's position as a leader in the future of automobile production.

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Case Study-02: Electric Vehicle Battery Pack Assembly

The Challenge:

Leading the way in zero-emission electric vehicle solutions that have the potential to significantly improve cities all around the world is Lion Electric. The company understood that it had to immediately scale up its manufacturing capacity to satisfy the demand as the demand for its products skyrocketed. This required making certain that all aspects of its implementation—the suppliers, the product, and the process—were completed correctly the first time. Lion Electric contacted JR Automation to find a solution.

Solutions:

The Lion and JR Automation teams realized they would need to first conceptualize the end-to-end solution because of the tight implementation timeframe. To do this, JR Automation constructed a functional prototype system utilizing essential automation components needed for the final production system. This required developing a solution that could be modified as the project's scope changed and that could do Hi-Pot testing, adhesive application, pressing, lasers, welding, and final testing. The teams quickly scaled up the system to produce a fully automated solution after the prototype system started producing battery packs of production-grade. This process took only a few months.

The entire manufacturing system, which spans several JR Automation production sites, is a shining illustration of JR Automation's capacity to create intricate, substantial solutions for its clients.

Outcomes and Benefits:

Together, the two teams were able to forge solid cooperation and swiftly and efficiently develop a first-rate automation solution. The early involvement of JR Automation in the development phase provided the Lion Electric team with valuable insights into the optimal design and production of the product. By working together, the teams were able to develop a solution and steer clear of later, expensive changes.

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Conclusion

Automation and industrial robots revolutionize manufacturing by increasing accuracy and efficiency. These sophisticated machines carry out operations like welding and assembly with efficiency thanks to their intelligent controllers and sturdy mechanical constructions. They collaborate seamlessly with people, using sensors to gather information about their surroundings and guarantee safety. Automation technologies, such as intelligent systems, simplify procedures, lower mistake rates, and increase output. These developments transform manufacturing in the age of Industry 4.0, establishing "smart factories" where equipment works together harmoniously.

These developments optimize the potential of automation, from accurate robot programming to smooth integration into current systems. Respecting safety regulations guarantees productivity as well as people's health in a variety of industrial environments. Automation and industrial robots are essentially driving industries toward a future with increased safety, efficiency, and connectivity.